



The Indian international School, DSO

Chemistry Department

PRACTICAL

Systematic Quantitative Analysis (Volumetric Analysis)

KMnO₄ Vs H₂C₂O₄

OBJECTIVE: To prepare 100 ml of M/20 oxalic acid solution and use this solution to determine the molarity and strength of given KMnO₄

REQUIRMENTS: Conical flask, burette, pipette, burette stand, tripod stand, test tube, test tube stands.

INDICATOR: Potassium permanganate is self-indicator.

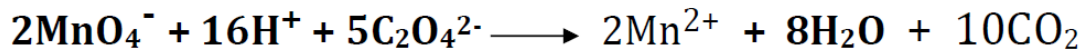
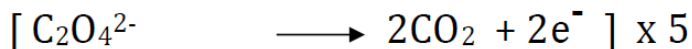
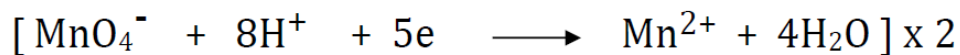
END POINT: Colorless to stable pink.

THEORY: Potassium permanganate is a strong oxidizing agent which oxidizes oxalic acid into carbon dioxide.

Molecular Equation



Ionic Equation



Weight calculation of oxalic acid

$$\begin{aligned}\text{mass of OXALIC ACID} &= \frac{\text{molarity} \times \text{molecular mass} \times \text{volume(in ml)}}{1000} \\ W_B &= \frac{M \times M_B \times V(\text{ml})}{1000} \\ &= \frac{1}{20} \times \frac{126 \times 100}{1000} \\ &= 0.63 \text{ gm}\end{aligned}$$

PROCEDURE

PREPARATION OF M/20 OXALIC ACID SOLUTION

Take 0.63g of oxalic acid in the standard flask. Then add water to make up the solution to 100ml. Shake the solution to get a homogenous solution.

TITRATION: KMnO_4 Vs $\text{H}_2\text{C}_2\text{O}_4$

- (1) Rinse and fill the burette with KMnO_4 solution and remove air bubble from the nozzle.
- (2) Suck out 10 ml of oxalic acid by using pipette and transfer it into 100 ml conical flask. Add about 10-15 ml of H_2SO_4
- (3) Heat the solution for 1-2 min then place the conical flask below burette.
- (4) Add KMnO_4 into conical flask drop by drop and stir well.
- (5) At first solution turns pink which on shaking disappears. Repeat the process until the solution becomes stable pink. Note down the reading of the burette.
- (6) Repeat the process to get concordant readings.

OBSERVATION TABLE

S.No.	Volume of (in ml) $\text{H}_2\text{C}_2\text{O}_4$	Burette reading		Volume of KMnO_4 (in ml) (Final-Initial)
		Initial	Final	
1				
2				
3				
4				
5				

concordant reading = ---- ml

Concordant reading = ---- ml of KMnO_4

CALCULATION:

$$\frac{M_{\text{KMnO}_4} \times V_{\text{KMnO}_4}}{M_{\text{oxalic acid}} \times V_{\text{oxalic acid}}} = \frac{2}{5}$$

$$M_{\text{KMnO}_4} = \frac{2 \times M_{\text{oxalic acid}} \times V_{\text{oxalic acid}}}{V_{\text{KMnO}_4} \times 5}$$

$$= \frac{2 \times 1/20 \times 20}{\text{---} \times 5}$$

$$= \text{-----}M$$

STRENGTH = Molarity x molar mass

= --- x 158

= ---- gL⁻¹

RESULT:

Molarity of KMnO₄= -----M

Strength of KMnO₄= -----g/L